Angad Basandrai

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Education

IIEST Shibpur — Howrah, West Bengal
Bachelor of Technology in Computer Science and Technology — *Aug. 2025 – Aug. 2029*

Experience

Research Intern
National Institute of Technology, Jalandhar — *oct. 2026 – present*

- researching multi-dimensional fatigue modeling through temporal feature extraction, cross-modal signal fusion, and physiological state estimation.

Research

Safe Opponent Exploitation in Colonel Blotto | *cs.GT* | *preprint* | *github* — sep. 2026 – oct. 2026

- developed an exact solver for Colonel Blotto games containing up to 4.3×10^{12} pure strategies using a double oracle framework to find safety-constrained exploitation strategies.
- evaluated exploitation strategies against **4,510 human allocations** from FiveThirtyEight Riddler Nation tournaments.

projects

qirc | *rust, quantum computing* | *github* | *writeup* — sep. 2026 – present

- qir compiler and built-in state-vector simulator
- removes **39.6%/36.3%** gates and CNOTs, comapred to **quarl**'s **35.2%/32,5%** on the nam suite.
- was placed third in **SegFault 2026**, at IISc Bengaluru.

syrix | *rust, opengl* | *github* — jul. 2026

- minimal browser from scratch
- working url navigation, hyperlink resolution, page reload, and home-page.

2d physics engine | *c++, opengl, glfw* | *github* — sep. 2025 – jul. 2026

- real-time 2d physics engine with rk4 numerical integration.
- supports **1,000+ bodies at 60+ fps** during runtime.

py-scorch | *python, cuda* | *github* — may 2026 – jun. 2026

- a neural network library in python with **cuda** support for gpu-accelerated computation.
- 182% training time compared to PyTorch on average.

Technical Skills

Languages: Rust, C++, C, Python, Java, JavaScript, SQL
Systems & Frameworks: LLVM, OpenGL, GLFW, React, PyTorch, PEFT
Tools: Git, Linux, Ghidra, Burp Suite

Side Quests

Founder and maintainer of **iiest.wiki**, including its associated sub-websites
Discovered issue in NTA websites, allowing brute forcing to **gain unauthorized account access** for JEE Mains, NEET and CUET. *more*
Found vulnerability in eScan CERT-In 7.2.0.70 allowing **bypass of its malware scan**. *more*
Rated **1000+** on Codeforces.com